Jiaming He

leanwithming.github.io/Homepage/

Email: JHE040@e.ntu.edu.sg
Mobile: +65 8419 8184

Research Interests

My research interests lie in **Trustworthy AI**, with a particular emphasis on **holistic red-teaming for advanced generative AI**. I focus on developing robust and reliable methods for deploying deep learning systems in the physical world. My work spans adversarial attacks, backdoor attacks, data poisoning (including unlearnable examples), and their mitigations. More recently, I have focused on the safety evaluation of agentic AI systems operating in both virtual and physical environments.

Education

- **Nanyang Technological University (NTU)** — Singapore
  Ph.D. Student; Funded by the NTU Research Scholarship — *2026 – Present*
- **Oxford Brookes University** — Oxford, United Kingdom
  B.Eng. in Software Engineering; Upper Second-Class Honours (2:1) — *Sep. 2021 – Jul. 2025*

Research Experience

- **Ant Group, Ant Research (CP Lab)** — China
  Research Intern — *May 2025 – Sep. 2025; May 2026 – Aug. 2026*
  – Studied personal-information extraction from video using large video-language models, advised by Dr. Cheng Hong, Dr. Zhicong Huang, and Prof. Neil Gong (Duke University).
- **University of Electronic Science and Technology of China** — Chengdu, China
  Research Assistant, Internet Data Security Laboratory — *Dec. 2024 – May 2025*
  – Researched foundation-model security under Prof. Hongwei Li (IEEE Fellow), including LLM jailbreaking and text-to-image backdoor attacks. First-authored work was accepted at CVPR, AAAI, ICME, and ICASSP.
- **King Abdullah University of Science and Technology** — Remote
  Research Intern — *May 2024 – Nov. 2024*
  – Studied machine unlearning and membership-inference attacks in vision-language models, including CLIP and LLaVA, advised by Prof. Di Wang.

Selected Publications and Preprint

[1] **TEAR: Temporal-aware Automated Red-teaming for Text-to-Video Models**
Jiaming He, Guanyu Hou, Hongwei Li, Zhicong Huang, Kangjie Chen, Yi Yu, Wenbo Jiang, Guowen Xu, Tianwei Zhang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. ***Oral (Top 0.7%).***

[2] **Watch Out for Your Guidance on Generation! Exploring Conditional Backdoor Attacks against Large Language Models**
Jiaming He, Wenbo Jiang, Hongwei Li, Guanyu Hou, Wenshu Fan, Rui Zhang
AAAI Conference on Artificial Intelligence, 2025. ***Oral (Top 5%).***

[3] **UTOPIA: Unlearnable Tabular Data via Decoupled Shortcut Embedding**
Jiaming He, Fuming Luo, Hongwei Li, Wenbo Jiang, Wenshu Fan, Zhenbo Shi, Xudong Jiang, Yi Yu
NeurIPS 2026 submission: reviewer scores 5/5/5 (5 = Accept); meta-review: Accept. Administratively desk-rejected for a reason unrelated to the technical evaluation.

[4] **PRESS: Defending Privacy in Retrieval-Augmented Generation via Embedding Space Shifting**
Jiaming He, Cheng Liu, Guanyu Hou, Wenbo Jiang, Jiacheng Li
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2025.

[5] **Weaponizing Tokens: Backdooring Text-to-Image Generation via Token Remapping**
Jiaming He, Wenbo Jiang, Guanyu Hou, Qiyang Song, Hongwei Li
IEEE International Conference on Multimedia & Expo (ICME), 2025. ***Oral.***

[6] **ChronoLock: Protecting Videos from Unauthorized Text-to-Video Personalization**
Jiaming He, J. Zhang, Guanyu Hou, S. Ye, H. Zhu, Yi Yu, X. Jiang
arXiv preprint arXiv:2606.21146, 2026.

[7] **Indistinguishable Agents: Training Agent Privacy as Twin-World Indistinguishability**
Jiaming He, Guanyu Hou, Yi Yu, Xudong Jiang

[8] **Evaluating Robustness of Large Audio Language Models to Audio Injection: An Empirical Study**
Guanyu Hou*, **Jiaming He***, Yinhang Zhou, Ji Guo, Yitong Qiao, Rui Zhang, Wenbo Jiang
*Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025. *Equal contribution.*

[9] **I2I Backdoor: Backdoor Attacks Against Image-to-Image Tasks**
Wenbo Jiang, Hongwei Li, **Jiaming He**, Rui Zhang, Guowen Xu, Tianwei Zhang, Rongxing Lu
IEEE Transactions on Dependable and Secure Computing (TDSC), 2025.

[10] **Whisper in the Folds: Towards Naturalness-aware Backdoor Attacks on Protein Representation**
Jiaming He, Peijia Qin, Hongwei Li, Guanyu Hou, Shenghao Wu, Wenbo Jiang, Yi Yu, Alex C. Kot
Preprint, 2025.

[11] **Measuring and Harnessing Video Language Models for Automated Personal Information Extraction**
Jiaming He, Zhicong Huang, Guanyu Hou, Kunsheng Tang, Peigui Qi, Cheng Hong, Neil Gong
Preprint, 2025.

[12] **Semantic Router: On the Feasibility of Hijacking MLLMs via a Single Adversarial Perturbation**
Changyue Li, Jiaying Li, Youliang Yuan, **Jiaming He**, Zhicong Huang, Pinjia He
International Conference on Machine Learning (ICML), 2026.

Technical Skills

- **Programming**: Python, C++, Java; PyTorch, TensorFlow **Tools**: LaTeX, Microsoft Office Suite **Language**: English